

Arnab Chakraborty

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Education

Gauhati University, B.Tech. in Information Technology

August 2023 - Expected 2027

Experience

Open-Source Contributor | MetaBrainz Picard

- Implemented a **long-awaited feature** to display both old and new file paths when renaming, **reducing user confusion by 30%** for file-naming scripts. (PR #2563)
- Developed a **custom titlization function** to **fixing 100+ instances** of incorrect capitalization, replacing Python's built-in method. Fixed an issue where "Children'S Music" was incorrectly capitalized, ensuring consistent metadata formatting. (PR #2548)
- Wrote **104 lines of Python code**, including **unit tests** using Pytest, **enhancing test coverage by 10%**, and ensuring robustness.
- **Technologies:** Git, GitHub, Python, PyQt, Pytest

Projects

Compress - 2026

compress.zytact.com

- Built privacy-first in-browser image compressor using WASM
- Implemented Lanczos resampling and size targeting
- No server uploads; 100% client-side processing
- **Technologies:** Rust, WebAssembly, TypeScript

Strokeshare - 2025

strokeshare.zytact.com

- Built infinite-canvas drawing app with export, undo/redo, and persistence
- Implemented performant state management using Zustand
- **Technologies:** Next.js, TypeScript, Tailwind CSS, React Konva, Zustand

InvenTrack - 2025

inventrack.zytact.com

- Developed full-stack inventory system with AI forecasting
- Designed multi-tenant database with Drizzle ORM
- Integrated LLM-powered recommendations
- **Technologies:** Next.js, Clerk, TypeScript, Tailwind CSS, Shadcn UI, TanStack Query, Drizzle ORM, Gemini API

Exiftuner - 2024

exiftuner.zytact.com

- Created batch EXIF editor with client-side processing
- Implemented metadata editing for GPS, timestamps, camera data
- **Technologies:** Next.js, Piexifjs, TypeScript, Tailwind CSS

Technical Skills

Programming Languages: C, Python, HTML, CSS, JavaScript, TypeScript

Fullstack Web Development: Node.js, Express.js, React.js, Next.js, Tailwind CSS, TanStack Query, Zustand, Drizzle ORM, RESTful APIs, tRPC, Auth.js, Better Auth, Clerk

Database: MongoDB, PostgreSQL

Tools

Git, GitHub Actions, Vim/NeoVim, Linux, Jest, Vitest, Pytest, ESLint, Prettier